Question 1 : develop a java program to prepare a simulated data set with unique instances

Program:

```
import java.util.*;

public class DatasetCreation {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter No. of attributes: ");
        int nAttr = sc.nextInt();
        sc.nextLine();

        ArrayList<String> attrs = new ArrayList<>();
        System.out.println("Enter Attribute names:");
        for (int i = 0; i < nAttr; i++) {
            attrs.add(sc.nextLine());
        }

        System.out.print("Enter No. of instances: ");
        int nInst = sc.nextInt();

        ArrayList<String> dataset = new ArrayList<>();

        int count = 1;
        while (dataset.size() < nInst) {
            System.out.println("Enter data for Instance " + count + ":");
            String data = "";
            for (int i = 0; i < nAttr; i++) {
                String val = sc.next();
                data += val + "\t";
            }
```

```
            if (!dataset.contains(data)) {
                dataset.add(data);
                count++;
            } else {
                System.out.println("Duplicate instance detected! Please re-enter unique data.");
            }
        }

        System.out.println("\nDataset:");
        for (String attr : attrs) {
            System.out.print(attr + "\t");
        }
        System.out.println();

        for (String d : dataset) {
            System.out.println(d);
        }
        sc.close();
    }
}
```

Q 2)develop a program to calculte the chi square value using python

program:

```
n = int(input("Enter number of values: "))

observed = []
expected = []

print("Enter observed frequencies:")
for i in range(n):
    observed.append(int(input()))

print("Enter expected frequencies:")
for i in range(n):
    expected.append(int(input()))

chi = 0

for i in range(n):
    chi += (observed[i] - expected[i]) ** 2 / expected[i]

print("Chi-Square Value:", chi)
```

Q 3)develop a progra, of navie bayesian classifiation using python

program:

```
from sklearn.naive_bayes import GaussianNB

n = int(input("Enter number of training records: "))

X = []
y = []

for i in range(n):
    a = float(input("Enter first value: "))
    b = float(input("Enter second value: "))
    c = int(input("Enter class (0 or 1): "))

    X.append([a, b])
    y.append(c)

model = GaussianNB()
model.fit(X, y)

a = float(input("Enter test first value: "))
b = float(input("Enter test second value: "))

result = model.predict([[a, b]])

print("Predicted Class:", result[0])
```

Q 4)develop a python programs on the k-meaan algorithm

program :

```
from sklearn.cluster import KMeans

n = int(input("Enter number of data points: "))

X = []
for i in range(n):
    a = float(input("Enter X value: "))
    b = float(input("Enter Y value: "))
    X.append([a, b])

print("\nBefore Clustering:")
for i in range(n):
    print(X[i])

k = int(input("\nEnter number of clusters: "))

model = KMeans(n_clusters=k, random_state=0, n_init=10)
model.fit(X)

print("\nAfter Clustering:")

for i in range(k):
    print("\nCluster", i + 1)
    for j in range(n):
        if model.labels_[j] == i:
            print(X[j])

print("\nCluster Centers:")
print(model.cluster_centers_)
```

q 5)develop a java programs on the k-meaan algorithm

program :

```
import java.util.*;

class KMeans {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        System.out.print("Enter number of points: ");
        int n = sc.nextInt();

        double[][] a = new double[n][2];

        for (int i = 0; i < n; i++) {
            System.out.print("Enter X and Y values: ");
            a[i][0] = sc.nextDouble();
            a[i][1] = sc.nextDouble();
        }

        System.out.print("Enter number of clusters: ");
        int k = sc.nextInt();

        double[][] c = new double[k][2];

        for (int i = 0; i < k; i++) {
            c[i] = a[i].clone();
        }

        int[] label = new int[n];
```

```
for (int t = 0; t < 10; t++) {
    for (int i = 0; i < n; i++) {
        double min = Double.MAX_VALUE;

        for (int j = 0; j < k; j++) {
            double d = Math.pow(a[i][0]-c[j][0], 2)
                    + Math.pow(a[i][1]-c[j][1], 2);

            if (d < min) {
                min = d;
                label[i] = j;
            }
        }
    }

    double[][] sum = new double[k][2];
    int[] count = new int[k];

    for (int i = 0; i < n; i++) {
        int j = label[i];
        sum[j][0] += a[i][0];
        sum[j][1] += a[i][1];
        count[j]++;
    }

    for (int j = 0; j < k; j++) {
        if (count[j] > 0) {
            c[j][0] = sum[j][0] / count[j];
            c[j][1] = sum[j][1] / count[j];
        }
    }
```

```
        }

        System.out.println("Cluster Results:");

        for (int i = 0; i < n; i++)
            System.out.println(Arrays.toString(a[i]) +
                    " -> Cluster " + (label[i] + 1));

        System.out.println("Cluster Centers:");

        for (int i = 0; i < k; i++)
            System.out.println(Arrays.toString(c[i]));

        sc.close();
    }
}
```

Q 6)delevelop a python program on apriori algorithm generate the frequent item setss and association rules

program:

```
import pandas as pd
from mlxtend.preprocessing import TransactionEncoder
from mlxtend.frequent_patterns import *

n = int(input("Enter number of transactions: "))
supp = float(input("Enter minimum support: "))
conf = float(input("Enter minimum confidence: "))

tx = []

for i in range(n):
    tx.append(input("Enter transaction: ").split(","))

te = TransactionEncoder()
df = pd.DataFrame(te.fit(tx).transform(tx), columns=te.columns_)

freq = apriori(df, min_support=supp, use_colnames=True)
rules = association_rules(freq, metric="confidence", min_threshold=conf)

print("\nFrequent Itemsets:")
print(freq)

print("\nAssociation Rules:")
print(rules[["antecedents", "consequents", "support", "confidence"]])
```

Q 7) develop a program to compute display dissimilrrity matrix (for your own data set containing at least four insstance with two attributes) using python

program:

```
import numpy as np
from scipy.spatial.distance import cdist

n = int(input("Enter number of instances: "))

data = []

for i in range(n):
    a = float(input("Enter first attribute: "))
    b = float(input("Enter second attribute: "))
    data.append([a, b])

d = cdist(data, data, metric='euclidean')

print("\nDissimilarity Matrix:")
print(d)
```

Q)Write a Python program to take a dataset dynamically from the user and visualize it using Histogram, Box Plot, Bar Chart, and Pie Chart.

program:

```
import matplotlib.pyplot as plt

n = int(input("Enter number of students: "))

names = []
marks = []

for i in range(n):
    name = input("Enter student name: ")
    mark = int(input("Enter marks: "))

    names.append(name)
    marks.append(mark)

# Histogram
plt.hist(marks)
plt.title("Histogram")
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.show()

# Box Plot
plt.boxplot(marks)
plt.title("Box Plot")
plt.ylabel("Marks")
plt.show()
```

```
# Bar Chart
plt.bar(names, marks)
plt.title("Bar Chart")
plt.xlabel("Students")
plt.ylabel("Marks")
plt.show()

# Pie Chart
plt.pie(marks, labels=names, autopct='%1.1f%%')
plt.title("Pie Chart")
plt.show()
```